The Eye — a short guide

Virtual Optics Laboratory · topic “The Eye” · experiments 17, 18 and 19

This part of the laboratory treats the eye as an optical instrument: two refracting elements, the **cornea** and the **lens**, must bring light from an object to a focus exactly on the **retina**. If the focus lands in front of the retina or behind it, the image is blurred. Everything on the screen (rays, focus, blur, numbers) is calculated from the model described below.

1. The theory in brief

Vergence and power

Light going to, or coming from, a point at the distance *l* (in metres) has the **vergence** L = 1/*l*, measured in dioptres (D). Light spreading out from a real object is negative: an object 25 cm away sends L = −4 D; a distant object sends L = 0.

A curved surface between two media adds its **power** P to the vergence:

```
P = (n′ − n) / r          L′ = L + P
```

Inside the eye, in a medium of index n, the vergence is n/*l*. You set the **curvature**; the power follows from it: a more curved surface (smaller r) is stronger.

The model eye

It is the Gullstrand–Emsley schematic eye, a standard textbook model. The fluids in the eye have the index 4/3, the lens 1.416.

Part	Value when relaxed	Power
Cornea	radius 7.8 mm	42.74 D
Lens	radii 10 mm and 6 mm, 3.6 mm thick	21.75 D
Whole eye	length 23.90 mm	60.48 D

The cornea does about 70% of the focusing. The lens is the part that can change.

Accommodation and age

To see a near object the lens becomes more curved and therefore stronger. This is **accommodation**. The amount needed, in dioptres, is

```
A = K − L
```

where L is the vergence arriving at the cornea and K is the eye’s refractive error (0 for a normal eye). A normal eye looking at 25 cm needs 4 D.

The lens stiffens with age, so the largest accommodation falls. The laboratory uses Hofstetter’s average for the population:

```
A_max = 18.5 − 0.30 × age    (D, age in years)
```

Age	A_max	Near point of a normal eye
20	12.5 D	8 cm
45	5.0 D	20 cm
55	2.0 D	50 cm
62 and older	0 D	no near focus

The **far point** is the farthest point seen sharply (relaxed lens); the **near point** is the nearest (lens fully accommodated). Near point distance = 1/(A_max − K).

Vision problems

Condition	Cause in the model	What happens with a distant object	Correction
Normal	length 23.90 mm, K = 0	focus on the retina, lens relaxed	none
Myopia (short sight)	eye too long: 25.04 mm, K = −3 D	focus 1.14 mm *in front of* the retina; far point only 33.3 cm away	negative (concave) lens
Hyperopia (long sight)	eye too short: 22.85 mm, K = +3 D	focus 1.05 mm *behind* the retina unless the eye accommodates	positive (convex) lens
Presbyopia	normal eye, age 55: lens too stiff	distance is sharp; the near point has moved out to 50 cm	positive reading glasses

About 1 mm of extra length makes the eye roughly 2.6 D short-sighted. The same errors can also come from the cornea: a radius of 7.29 mm instead of 7.8 mm gives about −3 D on a normal-length eye.

Glasses

Glasses are a thin lens of power F_S at the **vertex distance** d in front of the cornea (12 mm unless you change it). They change the vergence, and the light then travels the distance d before it meets the eye:

```
L′ = L + Fs          L_cornea = L′ / (1 − d·L′)
```

The lens that fully corrects an error K for distance is

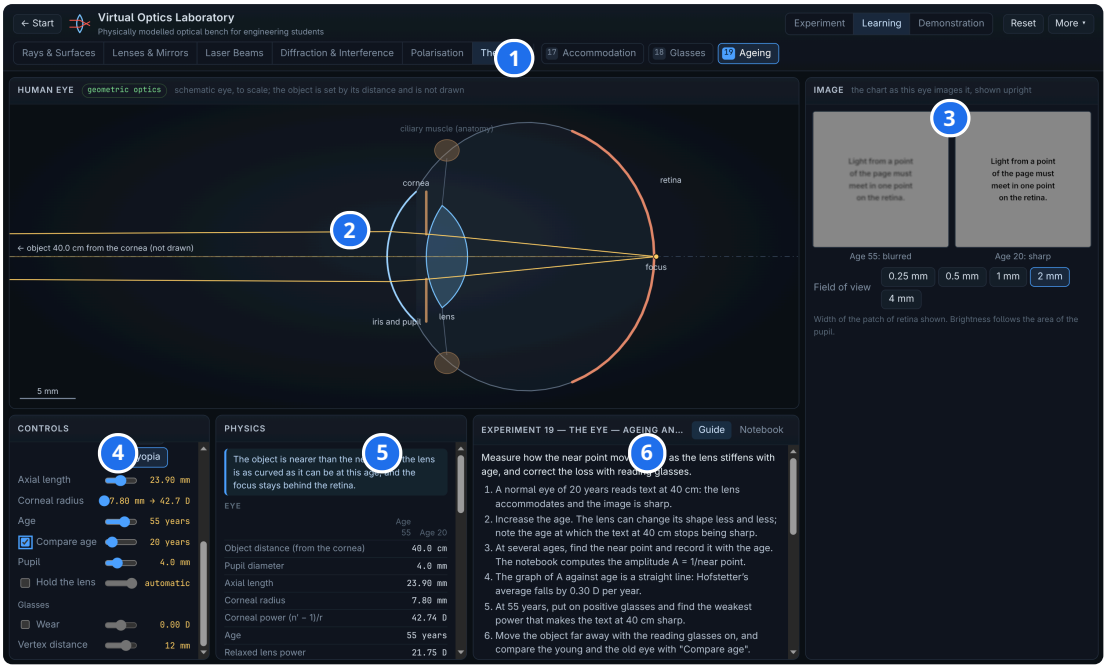
```
Fs = K / (1 + d·K)
```

Blur

Each point of the object sends a cone of light through the pupil. If the cone does not come to a point on the retina, it lights a small disc there: the **blur disc**. Its diameter grows in proportion to the **pupil diameter** and, very nearly, to the **focus error in dioptres**: about 75 µm per dioptre for a 4 mm pupil. The image panel shows the chart at its true size on the retina, smeared by exactly that disc.

The **sharpness** word compares the blur disc with the stroke width of the smallest letters: up to ¼ of a stroke is “sharp”, up to 1 “slightly blurred”, up to 3 “blurred”, more is “unreadable”.

2. How to use it



The eye workspace, here in experiment 19 with a 55-year-old and a 20-year-old eye reading at 40 cm.

1 Topic and experiment. Choose "The Eye", then experiment 17, 18 or 19. The mode switch is at the top right.

2 The eye, to scale. Yellow lines are rays from the object; the dot is where they focus. A red bar on the retina marks the blur patch. The lens changes shape as the eye accommodates. The object is set by its distance and is not drawn.

3 Image. The chart as this eye images it. With glasses on, or with a second age, the pictures appear side by side. "Field of view" is only a zoom.

4 Controls. *Object:* distance (the left end of the slider is a distant object), letter size, chart. *Eye:* condition, axial length, corneal radius, age, pupil. *Hold the lens* stops the automatic accommodation. *Glasses:* wear, power in 0.25 D steps, vertex distance.

5 Physics. All calculated values, in columns when several eyes are shown.

6 Guide and Notebook. The steps of the experiment, and a table that turns your readings into a result. "Check against theory" then compares your result with the accepted value.

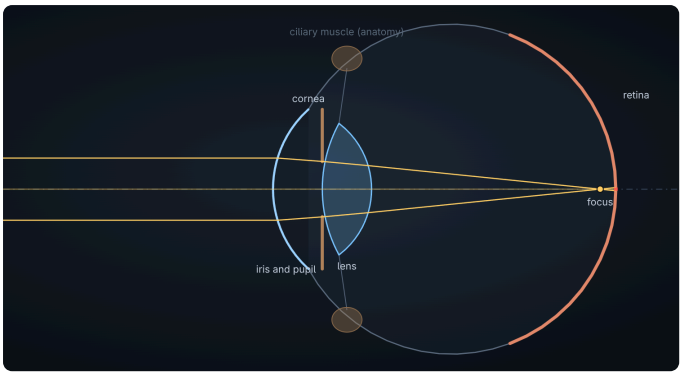
The three modes

Learning shows every value. **Experiment** hides the calculated ones ("to be determined") and the rays inside the eye, so you must work from the image; "Reveal theory" shows them afterwards, and "Use an unknown eye" gives you an eye whose error you have to find. **Demonstration** adds a "Sweep" button that moves one setting slowly back and forth.

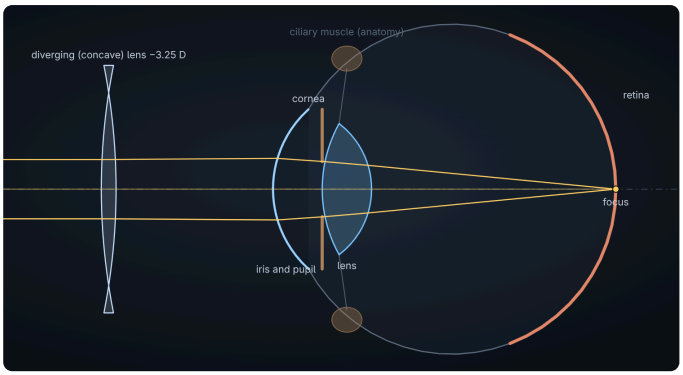
The three experiments

Experiment	What you do	What the notebook calculates
17 · Accommodation	Move the object closer and watch the lens thicken. Find the far point and the near point: the last distances at which the image is still sharp.	A = 1/near − 1/far (distances in metres; enter 0 for a distant far point)
18 · Glasses	Start with the short-sighted eye and a distant chart. Put on glasses and change the power until the chart is as sharp as it gets. Repeat with "Hyperopia" and the lens held relaxed.	K from the power you found, allowing for the vertex distance
19 · Ageing	Read text at 40 cm and raise the age until it blurs. Measure the near point at several ages. At 55, add positive glasses until the text is sharp again.	A = 1/near point for each age, and a graph whose slope should be −0.30 D per year

3. Why do I see this?



Short-sighted eye, no glasses. Light from a distant object focuses 1.14 mm in front of the retina and spreads again; the red bar is the 235 µm blur patch.



The same eye with −3.25 D glasses. The lens spreads the light slightly before it enters the eye, which moves the focus back onto the retina.

42.7 D + 21.8 D is 64.5 D. Why does the eye show 60.5 D?

Powers add up only for lenses in contact. The lens sits a few millimetres behind the cornea, and for separated elements the total is $P_1 + P_2 - (d/n)\cdot P_1 \cdot P_2$, which is smaller.

The long-sighted eye sees the distant chart sharply without glasses.

It is young, so it simply accommodates by 3 D ("Accommodation in use 3.00 D"). Tick *Hold the lens* and set it to relaxed: the focus drops 1.05 mm behind the retina and the chart is unreadable. This is why long sight often goes unnoticed in young people.

The −3 D eye needs −3.11 D glasses, not −3.00 D.

The glasses sit 12 mm in front of the eye. $F_S = K/(1 + dK) = -3/(1 - 0.036) = -3.11$ D. Set the vertex distance to 0 (a contact lens) and it becomes −3.00 D. For the +3 D eye the same formula gives +2.90 D.

With −3.00 D the chart is "slightly blurred", with −3.25 D it is "sharp".

Glasses come in 0.25 D steps, so the exact −3.11 D cannot be set. −3.00 D leaves 0.10 D uncorrected: with the letters the experiment starts with, an 8 µm blur disc against strokes of about 10 µm. −3.25 D is 0.13 D too strong, and the young eye hides that by accommodating 0.13 D.

Even −4 D looks sharp on the short-sighted eye.

For the same reason: a young eye accommodates away any over-correction. That is why the rule is to take the *weakest* negative lens that gives a sharp image.

Reading glasses make the distance blurred.

With +1.50 D the 55-year-old eye sees sharply only from 29 cm to 68 cm. Beyond 68 cm the eye would have to relax more than it can.

A smaller pupil makes the image sharper but darker.

The blur disc is proportional to the pupil diameter, the brightness to its area: half the pupil gives half the blur and a quarter of the light. The image is drawn at full brightness only for the largest pupil (8 mm), so it looks grey at 4 mm.

The short-sighted eye has a closer near point than the normal eye.

Its whole range is shifted towards the eye: near point = 1/(A_max − K). At 25 years (A_max = 11 D) that is 7.1 cm for K = −3 D, against 9.1 cm for a normal eye.

An unknown eye looks like a normal eye in the drawing.

On purpose. Its length and cornea are hidden, so it is drawn with the standard dimensions and labelled "not to scale"; otherwise the answer could be measured off the picture.

4. What the model leaves out

- **It is a simplified eye.** Astigmatism, lens aberrations, diffraction at the pupil and everything the retina and brain do with the image are not modelled. "Sharp" means "in geometric focus"; it is not a measurement of visual acuity.
- **The age formula is an average.** Real people of the same age differ by several dioptres. The formula reaches zero at 62, while real eyes keep about 1 D of apparent accommodation because of their depth of focus.
- **Only the lens ages here.** Age changes how far the lens can accommodate and nothing else; the cornea and the length of the eye are separate controls.
- **The image on the retina is upside down.** The panel shows it upright, as it is seen.

The numbers in this guide come from the laboratory's own model; the on-screen values quoted were read from the running application. Model: Gullstrand–Emsley schematic eye; accommodation by age after Hofstetter's average. The full description is in docs/PHYSICS.md, sections 16 and 17.